PAVLOS ZOUBOULOGLOU

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EDUCATION

Doctoral student, UNC Chapel Hill, Statistics and Operations Research — August 2019 - May 2024
Advisor: Amarjit Budhiraja

B.Sc. in Mathematics, University of Athens (NKUA), GPA: 8.50/10 — September 2014 - June 2019

Study Abroad in Mathematics, TU Berlin — April 2018 -August 2018

Study Abroad in Computational Mathematics, ERAU — August 2017 - December 2017

Minor in Political Science, Deree — September 2015 - December 2017

PROFESSIONAL EXPERIENCE

Postdoctoral fellow (Cluster of Excellence), University of Münster — August 2024 - Present
Institute for Mathematical Stochastics, Department of Applied Mathematics
Mentor: Chiranjib Mukherjee

PUBLICATIONS AND PREPRINTS

1. Marie-Christine Düker, **Pavlos Zoubouloglou.** *On the Breuer–Major Theorem on Hilbert spaces, functional time-series, and feature learning* Preprint (2026). [Preprint]
2. Marie-Christine Düker, **Pavlos Zoubouloglou.** *Breuer–Major Theorems for Hilbert Space-Valued Random Variables.* Preprint (2026). [arXiv]
3. Georgia Karali, Alexandra Stavrianidi, Konstantinos Tzirakis, **Pavlos Zoubouloglou.** *On the Density of the Supremum of Nonlinear SPDEs.* Preprint (2026). [arXiv]
4. Marie-Christine Düker, **Pavlos Zoubouloglou.** *The Fourth Moment Theorem on Hilbert Spaces.* Preprint (2025). [arXiv]
5. Marie-Christine Düker, **Pavlos Zoubouloglou.** *Scaling Limits for Sample Autocovariance Operators of Hilbert Space-Valued Linear Processes.* Preprint (2025). [arXiv]
6. Amarjit Budhiraja, Adam Waterbury, **Pavlos Zoubouloglou.** *Large Deviations for Empirical Measures of Self-Interacting Markov Chains. Stochastic Processes and their Applications* **186** (2025), 104640. [arXiv] [Journal]
7. Stefanos Kechagias, Vladas Pipiras, **Pavlos Zoubouloglou.** *Cyclical Long Memory: Decoupling, Modulation, and Modeling. Stochastic Processes and their Applications* **175** (2024), 104403. [arXiv] [Journal]
8. Amarjit Budhiraja, **Pavlos Zoubouloglou.** *Large Deviations for Small Noise Diffusions Over Long Time. Transactions of the American Mathematical Society, Series B* **11** (2024), 1–63. [arXiv] [Journal]
9. *__Pavlos Zoubouloglou__, Eduardo García Portugués, J. S. Marron. *Scaled Torus Principal Component Analysis. Journal of Computational and Graphical Statistics* **32**(3) (2023), 1024–1035. [arXiv] [Journal]
10. *Vladas Pipiras, **Pavlos Zoubouloglou**, Themistoklis Sapsis. *Cyclical Long Memory in Ship Motions at Non-Zero Speed.* Proceedings of the 18th International Ship Stability Workshop (2022), 297–306. [Preprint] [Journal]

* *indicates papers where author order reflects contribution; absence of* * *indicates alphabetical author order.*

AWARDS AND SCHOLARSHIPS

1. **Young Mathematicians Conference Network Fund**, Department of Mathematics Münster
   (in support of organizing the event "Summer School in Interacting Random Systems") — 20000 (in €)
2. **Internationalization Fund**, University of Münster
   (in support of organizing the event "Summer School in Interacting Random Systems") — 3200 (in €)

3. **Student Success Ambassador Award**, Graduate School (UNC)
4. **M. R. Leadbetter Graduate Student Excellence Award**, Dept. of Statistics and OR
5. **2023-24 Dissertation Completion Fellowship**, Graduate School, UNC 22000 (in $)
6. **2022 Summer Research Fellowship**, Graduate School, UNC 5000 (in $)
7. **Erasmus Study Abroad Scholarship**, Erasmus+ through NKUA, 2018 2250 (in €)
8. **Full Study Abroad Scholarship**, Stavros Niarchos Foundation and Deree, 2017 7950 (in €)
9. **Scholarship for Minor Certificate**, Stavros Niarchos Foundation, 2015 1400 (in €)

TEACHING EXPERIENCE

University of Münster

- Stochastic Analysis (2 times): Winter Semester 2025–26, Winter Semester 2024–25. Role: Teaching assistant, tutorial instructor, grader.
- Probability Theory I (1 time): Summer Semester 2025. Role: Tutorial instructor, grader.

University of North Carolina at Chapel Hill

- Introduction to Data Science (6 times): Fall 2021, Spring 2021, Fall 2020, Summer 2020, Spring 2020, Fall 2019. Role: Teaching assistang, lab instructor, grader.
- Comprehensive Written Exams Tutorials for Probability Theory and Statistical Theory (2 times): Summer 2023 and Summer 2021. Role: Instructor.
- Introduction to Data Models and Inference (2 times): Summer 2023, Summer 2021). Role: Teaching assistant, grader.
- Methods in Data Analysis (1 time): Summer 2020. Role: Teaching assistant.

OTHER SKILLS

Technical LaTeX, R, MATLAB
Languages Greek (Native), English (C2), German (A2-B1), Spanish (A2)

PROFESSIONAL ACTIVITIES

- Early Career Committee of the Cluster of Excellence, University of Münster May 2025 - Present
- Main organizer of a week-long event in Münster:
  "Summer School in Interacting Random Systems" June 2025
- STEM Pride of the Triangle Board Member, UNC August 2023 - May 2024
- STOR-Biostatistics student liaison, UNC June 2023 - May 2024
- Graduate Student Seminar Co-Organizer, UNC August 2021 - May 2024
- Graduate and Professional Student Federation Senator, UNC September 2021 - August 2022

REFEREEING

Bernoulli; Stochastic Analysis and Applications; Involve; Potential Analysis; Journal of the Royal Statistical Society: Series C (Applied Statistics); Journal of Statistical Planning and Inference; Proceedings of the Edinburgh Mathematical Society; ESAIM: Probability and Statistics.

INVITED TALKS

15th AIMS Conference University of Athens — Athens (GR), July 2026

Recent Advances in Time Series — Larnaca (CY), June 2026

19th Panhellenic Conference in Analysis NTUA — Athens (GR), December 2025

University of Münster, Stochastic Oberseminar — Münster (DE), November 2025

Friedrich-Alexander University — Erlangen (DE), June 2024

Università di Pisa — Pisa (IT), June 2023

One World Dynamics Seminar — Zoom, March 2023

UNC STOR Graduate Student Seminar — Chapel Hill (NC), October 2021 and September 2022

TRAININGS AND CONFERENCES WITH CONTRIBUTED TALK OR POSTER

Summer School in Applied Stochastic Analysis WIAS — Berlin (De), July 2026

3rd Congress of Greek Mathematicians University of Athens — Athens (GR), July 2026

53rd Probability Summer School Université Clermont Auvergne — Saint-Flour (FR), July 2025

World Congress in Probability and Statistics Bernoulli-IMS — Bochum (DE), August 2024

Summer School: PDEs and Randomness University of Maryland — College Park (MD), August 2024

Seminar on Stochastic Processes Rice University — Houston (TX), March 2024

Stochastic Analysis, Random Fields, and Applications, Michigan State University — East Lansing (MI), August 2023

Frontiers in Stochastic Analysis University of Illinois Chicago — Chicago (IL), August 2023

51st Probability Summer School Université Clermont Auvergne — Saint-Flour (FR), July 2023

Seminar on Stochastic Processes University of Arizona — Tucson (AZ), March 2023

Graduate Students Conference in Probability UW-Madison — Madison (WI), September 2022

Brazilian School of Probability — Campinas (BR), August 2022

TRAININGS AND CONFERENCES WITHOUT CONTRIBUTION

20 Years of the Fourth Moment Theorem University of Luxembourg — Luxembourg (LU), December 2025

Stochastic Analysis in Mathematics and Natural Sciences FORTH Institute — Crete (GR), June 2025

Spring School in Extrema of Logarithmically Correlated Random Fields Darmstadt Universität — Darmstadt (DE), March 2025

Northeast Probability Conference New York University — New York City (NY), November 2023

Southeastern Probability Conference I Duke University — Durham (NC), May 2023

Summer School in Lévy Processes NTUA — Athens (GR), July 2019